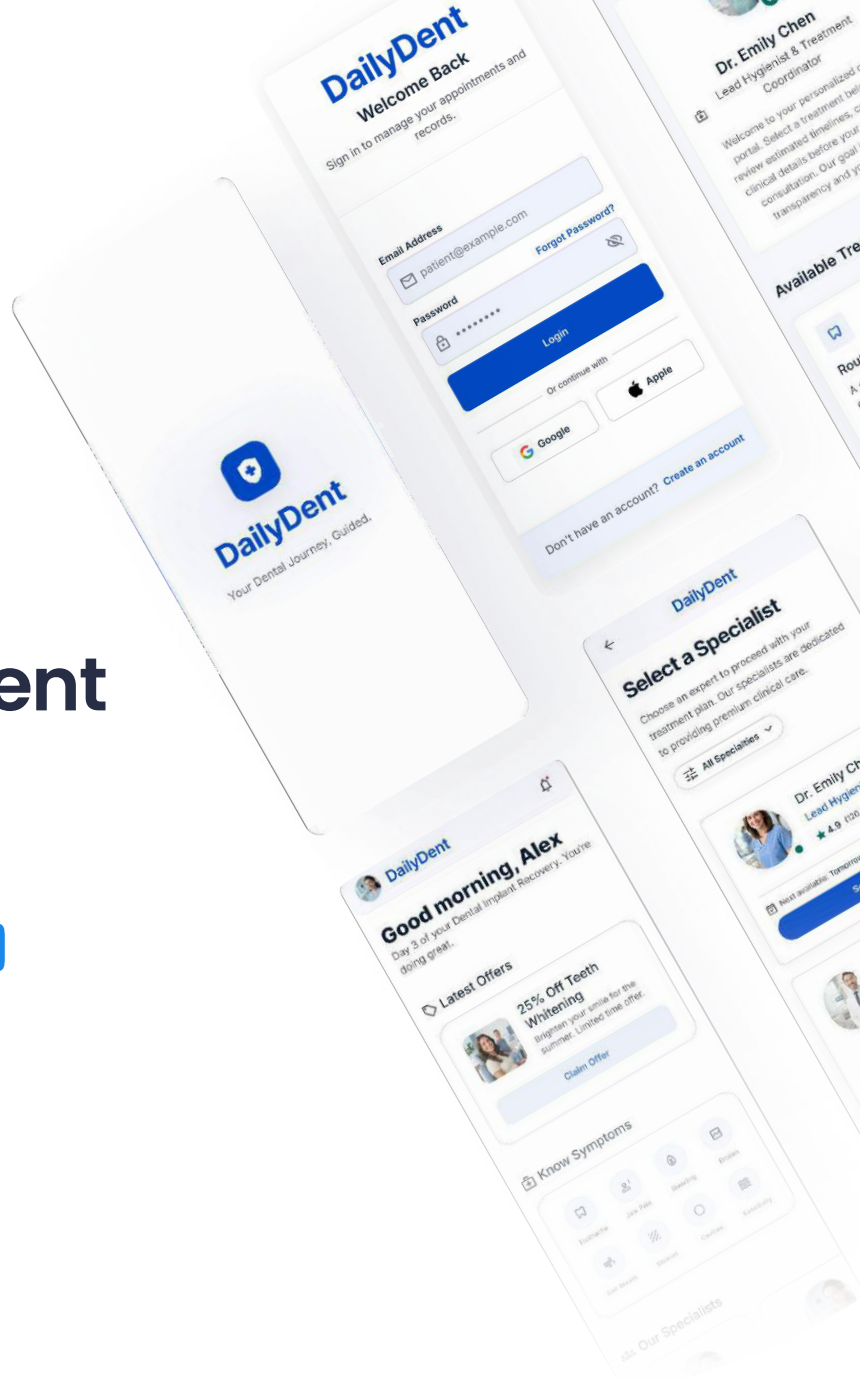
DailyDent

UX Case Study

Dental Treatment Case Study

Patient Engagement for Dental Treatments

K Balaji



Problem Statement

The dental industry faces a challenge in establishing a robust and engaging communication channel between patients and dentists. Patients are reluctant to dental treatments as they lack understanding of treatment procedures. Dentists find it difficult to aware patients about treatment plans and complexities. Sub-optimal post treatment care leads to- Decreased treatment compliance, heightened complications risk, and reduced patient satisfaction.

Objectives & Goals

- Enhance Patient awareness and Education
- Patient- Dentist real time Communication
- Treatment planning and collaboration for dentists
- Improve Post-treatment Compliance

UX Process

Discover

Define

Ideate

Design

Ideation

- Online booking and payment
- Providing a notification system for doctor appointments
- Providing symptoms articles to patient
- Using video calls in the examinations for some conditions
- Automatically adjusting patients medication dose schedule

Product Users

The application will be use by new patient / Existing patient and Doctors.

My Role

I had created to fill different parts of the design process and I was responsible to conduct all kinds of UX activities and there are some of the deliveries produced.

- Competitive analysis
- User persona
- User Journey Map
- Sketching
- Wireframes
- User flow
- Design
- Project Documentation

Competitive analysis

My Invisalign	DentalMonitoring	DentiCalc
Tracking	Remote Ortho	Clinical Education
Downloads > 1M	Downloads > 500K	Downloads 100K+
3.6/5.0	3.2/5.0	3.0/5.0

Feature Comparison Matrix

FEATURE	MY INVISALIGN	DENTALMONITORING	DENTICALC
Hardware Integration	✗	ScanBox Mount	✗
A.I. Analytics	✗	✓ Tooth Movement	✗
Visual Progress	Time-Lapse	AI Progress Reels	1000+ Animations
Doctor Messaging	✓	✓	✗
Gamification	✓ Wear-Time Logging	✓ Compliance	✗
Financial Tools	✗	✗	✓ Pricing Calculator

Customer Feedback & Sentiment Analysis

My Invisalign

Positive Trends
Patients appreciate the comfort of aligners and find the wear-time tracker highly motivating. The app successfully keeps users accountable to their treatment plan.

Negative Trends
Frustration over app stability. Users report the app freezing, unexpectedly signing them out, and in severe cases, wiping out their photo logs and treatment history.

DentalMonitoring

Positive Trends
Highly praised for convenience. Patients love seeing clear progress from home and avoiding weekly office visits. Parents find it incredibly easy for their children to use independently.

Negative Trends
Very few negative reviews exist, though historically some users report a slight learning curve when first using the physical ScanBox attachment to capture clear photos.

DentiCalc

Positive Trends
Dentists find it to be an excellent tool for patient education and case presentation. The vast library of animations effectively bridges the knowledge gap, leading to higher case acceptance.

Negative Trends
Some users encounter issues with animations failing to load. Additionally, international users have requested broader currency support and clearer instructions on how to customize pricing.

User Needs

- View available time slots on the calendar for easy selection. The calendar should get updated in real time for available slots.
- View available time slots on the calendar for easy selection. The calendar should get updated in real time for available slots.
- Automated confirmation emails/ SMS notifications for scheduled appointments. Also reminder notifications sent to patients closer to the appointment date.
- A cost calculator to estimate the total expenses of the treatment, including possible financing options

Features & Functionalities

To resolve user needs

Appointment Reminders & Confirmation

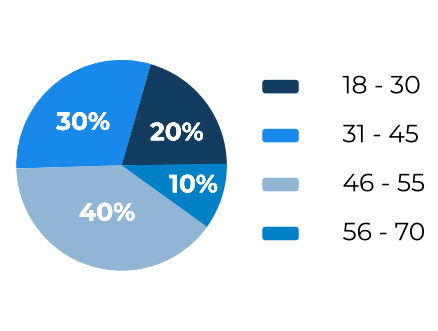
Treatment planning tools

Post-operative care guidelines

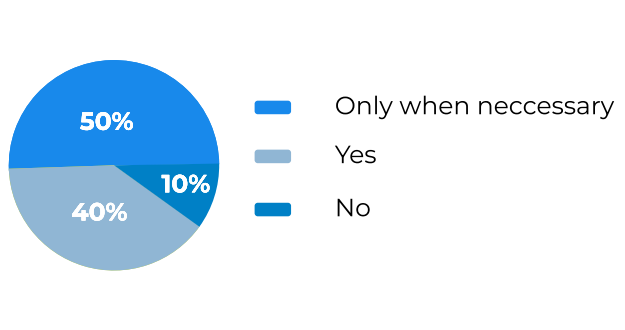
Patient Survey

We wanted to make sure that the user surveys are as simple and straight to the point as possible, we targeted average user as audience.

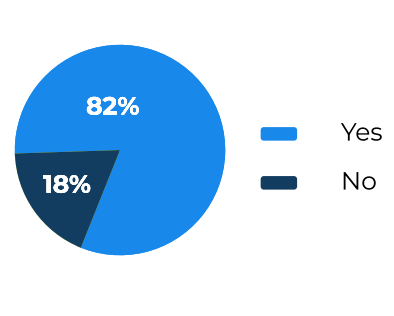
How old are you?



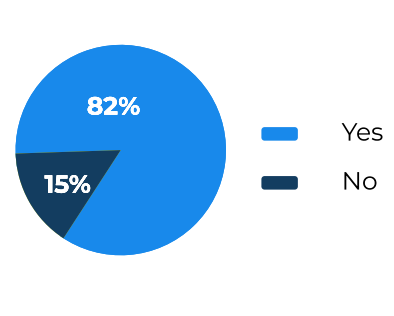
Do you regularly go to the clinc?



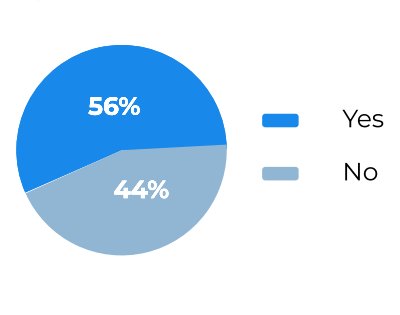
Do you suffer from toothache or symptoms ?



Have you ever booked an appointment with a Doctor online?



Do you prefer to use the online payment method?

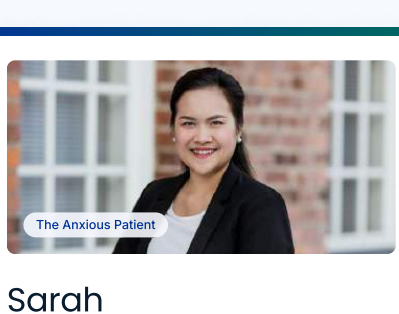


Unique Features

- Visual timeline of healing progress, allowing patients to compare and track changes over time
- Interactive tool that would visually demonstrate the treatment process and progress
- Calendar with reminders for follow-up appointments, medication schedules, and oral hygiene routines

Patient Profile

Understanding our users to build empathetic clinical experiences.



Sarah

Marketing Executive, 28

SCENARIO
Needs a dental implant, terrified of surgery and the recovery process.

PREFERRED PAYMENT
Apple Pay / Digital Wallet

PERSONALITY
Tech-savvy
Detail-oriented
Health-conscious
Anxious but Proactive

"I just want to know what's happening during the procedure so I can feel in control of my recovery."

Tech Fluency
High

Goals
- Understand every step of the procedure clearly.
- Feel 'in control' of the healing timeline.
- Reduce overarching anxiety regarding clinical visits.

Pain Points
- "I don't know what they're actually doing in there."
- Uncertainty about symptoms: "Is this swelling normal?"
- Overwhelmed by complex medical terminology.

Empathy Map

Says
"How long will the pain last?"
"I need to see a clear timeline of the recovery."

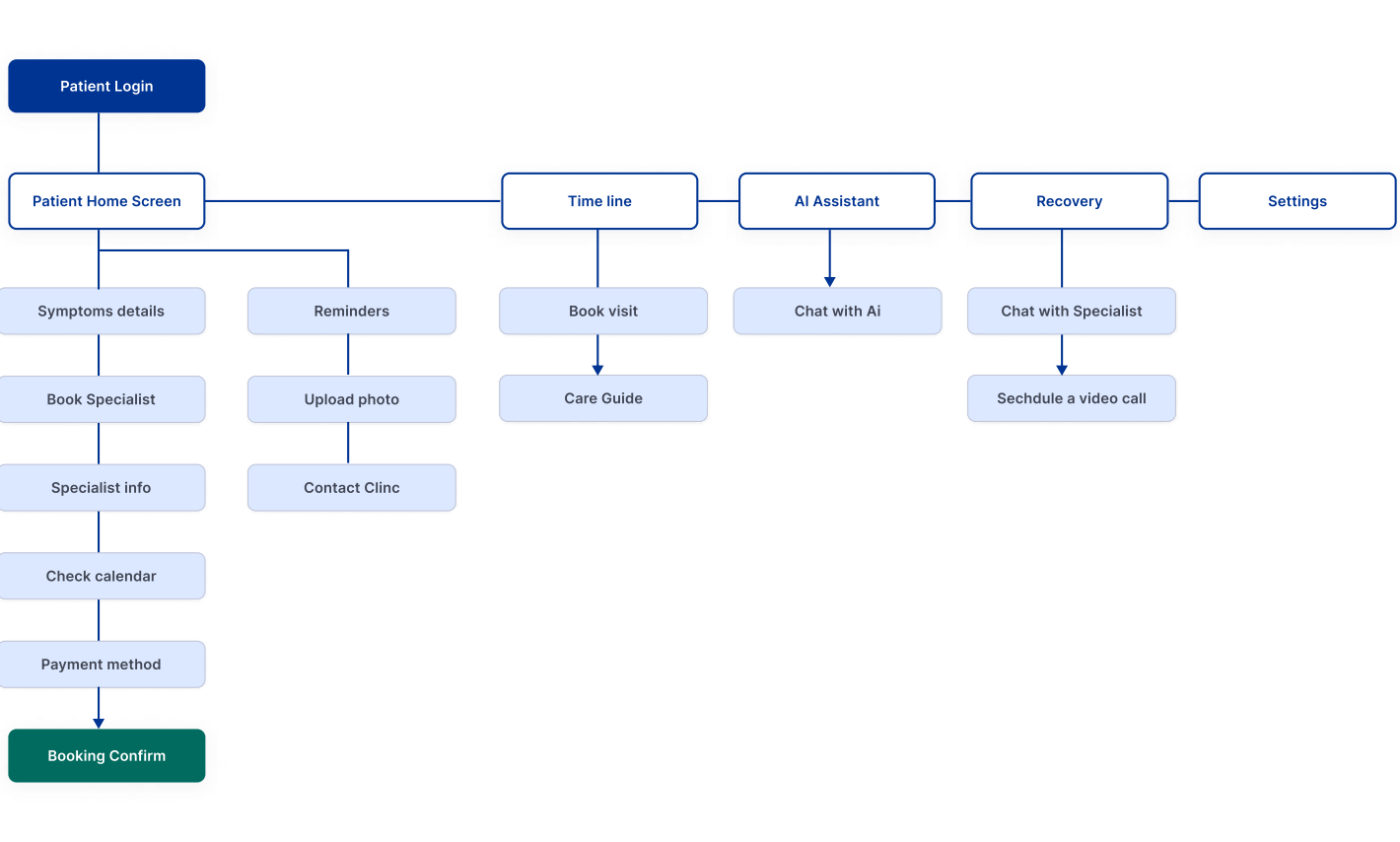
Thinks
"I hope I don't regret doing this procedure."
"Will my smile look natural afterward?"

Does
Searches symptoms online at 2 AM.
Double-checks medication times and sets strict alarms.

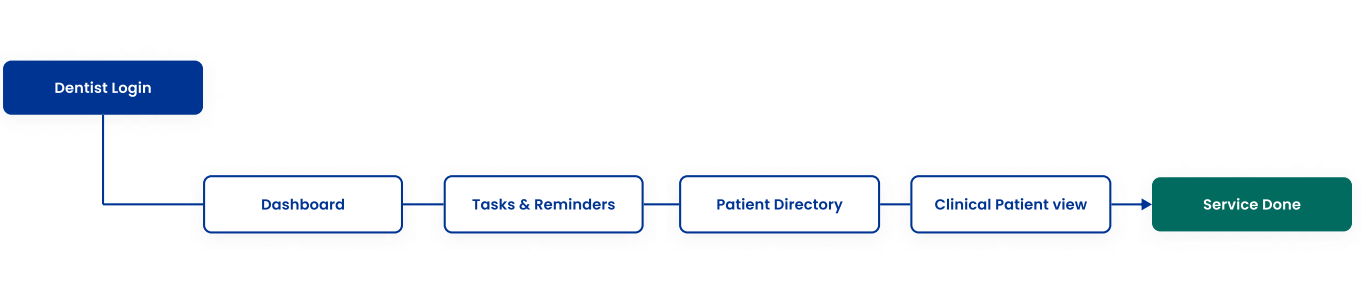
Feels
Anxious and highly vulnerable in the clinical chair.
Constantly seeking reassurance and validation from staff.

Patient Journey

User flow for scheduling dental appointments. A clean, high-trust clinical pathway.



Dental Specialist Journey



Design System

Brand Identity

DailyDent

The 'Soft-Touch' Interface

Rooted in "Modern Corporate" clinical serenity, the visual language is designed to evoke calm, reliability, and professional care. It prioritizes clarity and reassurance, acting as a supportive, quiet companion rather than a complex tool. Generous whitespace, sophisticated translucency, and high legibility define the patient experience.

Color Palette

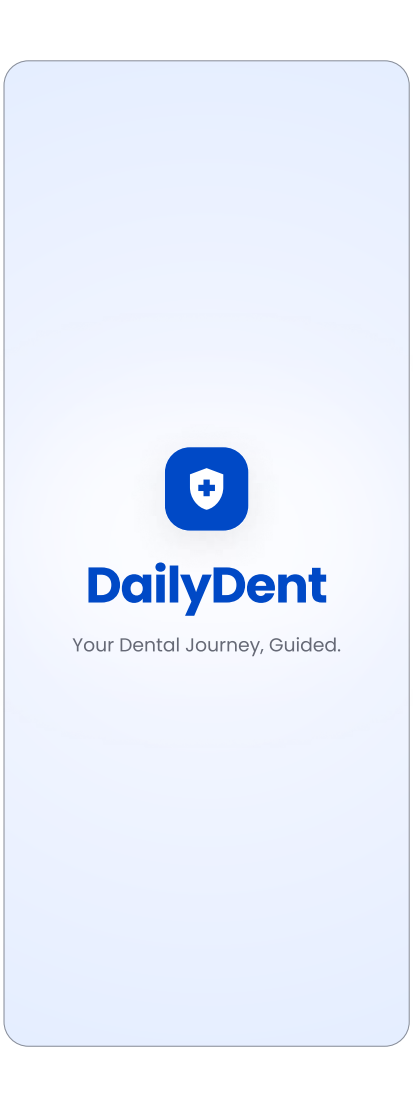
Primary Blue	Surface	Surface Container	Success / Healing	On-Surface
#2563eb	#f8f9ff	#eff4ff	#059669	#1a1c1e

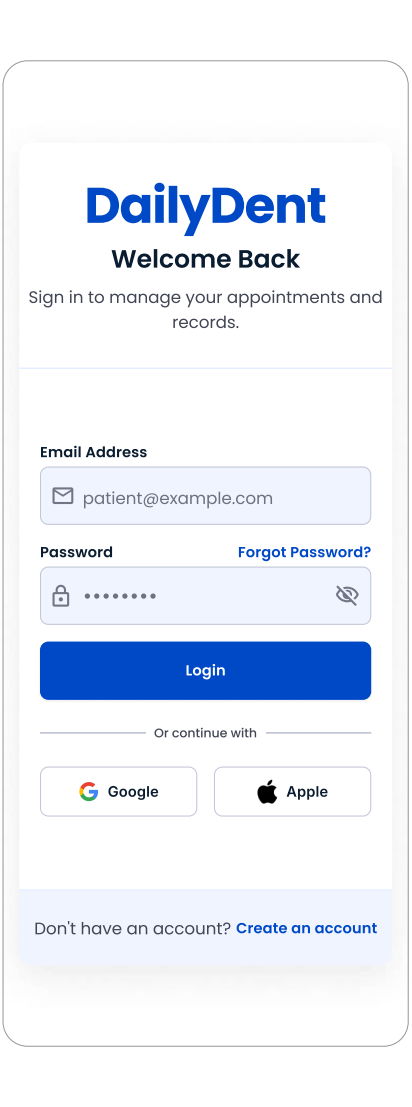
Typography — Poppins

DISPLAY	**Clinical Serenity** 48px / 56px / Bold / -2% tracking
HEADLINE	**A breathable layout** 32px / 40px / Semibold / -1% tracking
TITLE	Assisted Recovery Companion 20px / 28px / Medium
BODY	The technology feels like a supportive, quiet companion rather than a complex tool. Spacing follows an 8px base unit. Section-level gaps are aggressive to emphasize content grouping, while internal card padding remains generous to prevent information density from feeling overwhelming. 16px / 24px / Regular

Iconography

Home
Timeline
AI Assistant
Recovery
Settings
Notifications
Toothache
Jaw Pain
Bleeding
Clinical





DailyDent
Welcome Back
Sign in to manage your appointments and records.
Email Address: patient@example.com
Password: ••••••••
Forgot Password?
Login
Or continue with
Google
Apple
Don't have an account? Create an account

Profile

Promos

Symptoms articles

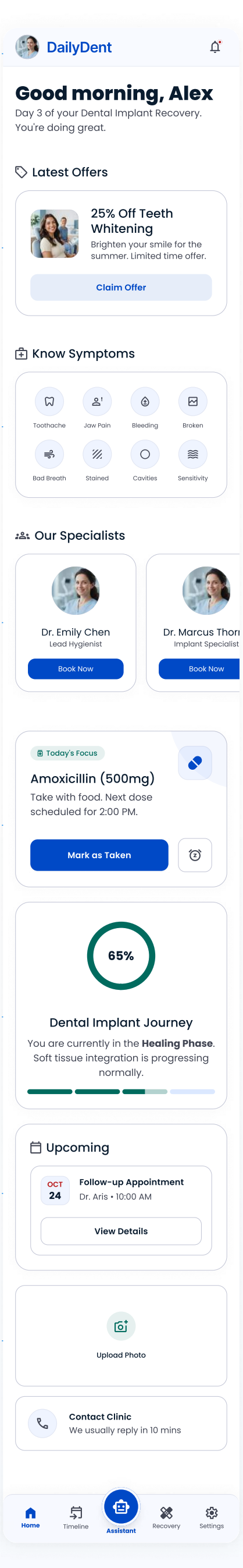
Specialist list

Specialist list

Patient healing journey

Patient appointments

Patients can upload dental pics



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Good morning, Alex
Day 3 of your Dental Implant Recovery. You're doing great.

Latest Offers
25% Off Teeth Whitening
Brighten your smile for the summer. Limited time offer.
Claim Offer

Know Symptoms
Toothache, Jaw Pain, Bleeding, Broken, Bad Breath, Stained, Cavities, Sensitivity

Our Specialists
Dr. Emily Chen, Lead Hygienist, Book Now
Dr. Marcus Thorn, Implant Specialist, Book Now

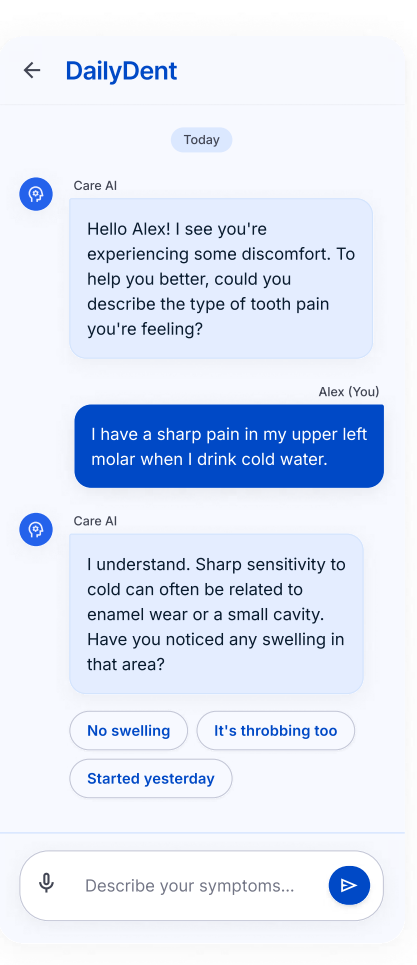
Today's Focus
Amoxicillin (500mg)
Take with food. Next dose scheduled for 2:00 PM.
Mark as Taken

65%
Dental Implant Journey
You are currently in the **Healing Phase**. Soft tissue integration is progressing normally.

Upcoming
OCT 24 Follow-up Appointment
Dr. Aris • 10:00 AM
View Details

Upload Photo

Contact Clinic
We usually reply in 10 mins



DailyDent
Today

Care AI: Hello Alex! I see you're experiencing some discomfort. To help you better, could you describe the type of tooth pain you're feeling?

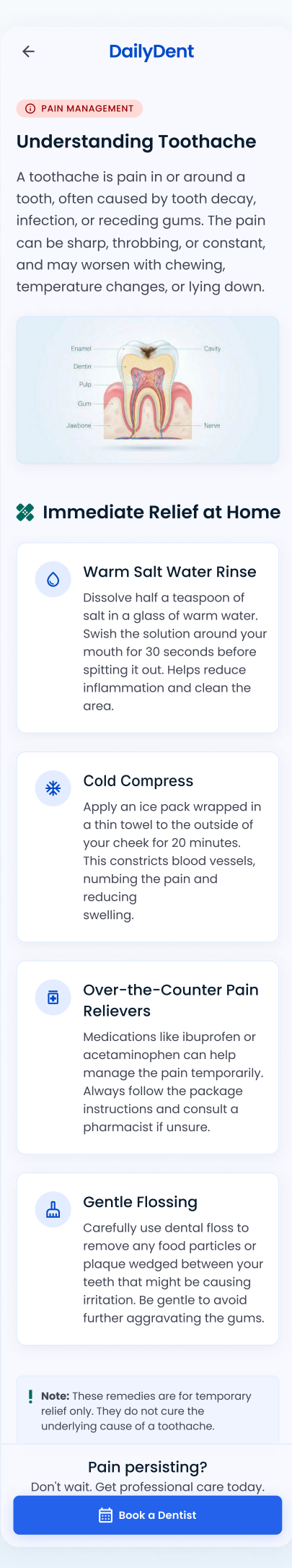
Alex (You): I have a sharp pain in my upper left molar when I drink cold water.

Care AI: I understand. Sharp sensitivity to cold can often be related to enamel wear or a small cavity. Have you noticed any swelling in that area?

No swelling
It's throbbing too
Started yesterday

Describe your symptoms...

Patient Home Screen



DailyDent

PAIN MANAGEMENT

Understanding Toothache

A toothache is pain in or around a tooth, often caused by tooth decay, infection, or receding gums. The pain can be sharp, throbbing, or constant, and may worsen with chewing, temperature changes, or lying down.

Immediate Relief at Home

Warm Salt Water Rinse
Dissolve half a teaspoon of salt in a glass of warm water. Swish the solution around your mouth for 30 seconds before spitting it out. Helps reduce inflammation and clean the area.

Cold Compress
Apply an ice pack wrapped in a thin towel to the outside of your cheek for 20 minutes. This constricts blood vessels, numbing the pain and reducing swelling.

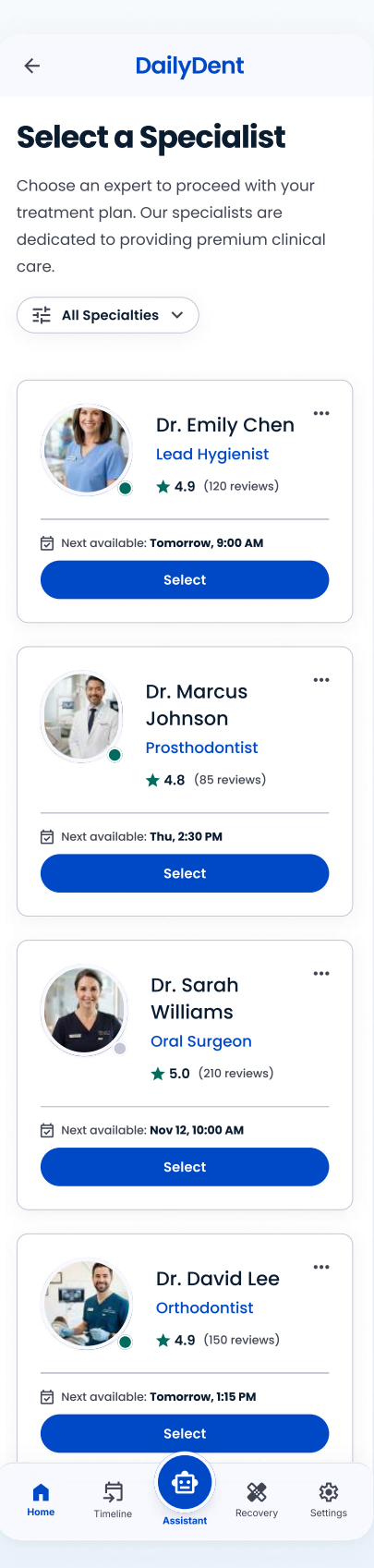
Over-the-Counter Pain Relievers
Medications like ibuprofen or acetaminophen can help manage the pain temporarily. Always follow the package instructions and consult a pharmacist if unsure.

Gentle Flossing
Carefully use dental floss to remove any food particles or plaque wedged between your teeth that might be causing irritation. Be gentle to avoid further aggravating the gums.

Note: These remedies are for temporary relief only. They do not cure the underlying cause of a toothache.

Pain persisting?
Don't wait. Get professional care today.
Book a Dentist

Symptoms article



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Select a Specialist
Choose an expert to proceed with your treatment plan. Our specialists are dedicated to providing premium clinical care.

All Specialties

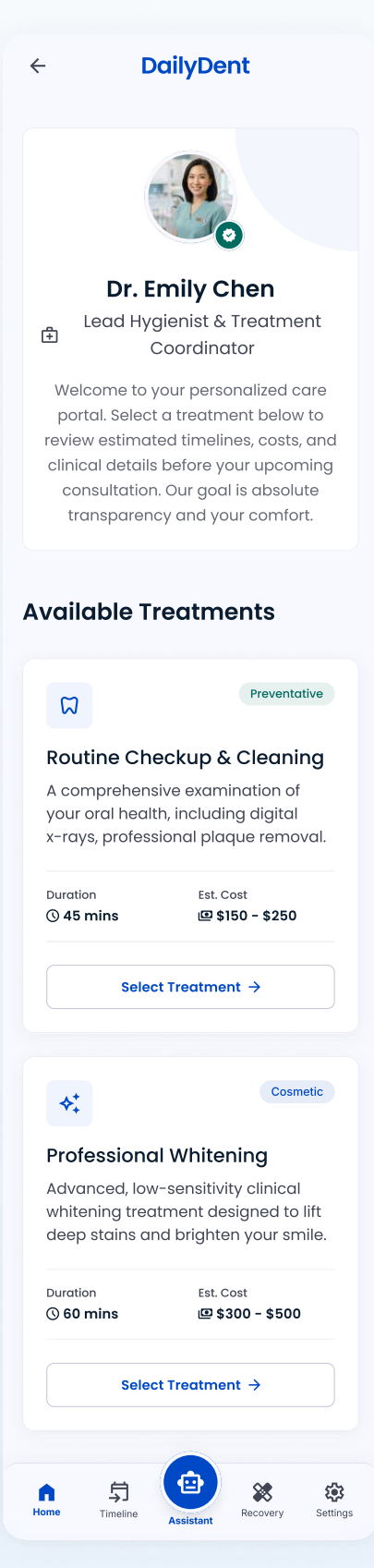
Dr. Emily Chen, Lead Hygienist, 4.9 (120 reviews)
Next available: Tomorrow, 9:00 AM
Select

Dr. Marcus Johnson, Prosthodontist, 4.8 (85 reviews)
Next available: Thu, 2:30 PM
Select

Dr. Sarah Williams, Oral Surgeon, 5.0 (210 reviews)
Next available: Nov 12, 10:00 AM
Select

Dr. David Lee, Orthodontist, 4.9 (150 reviews)
Next available: Tomorrow, 1:15 PM
Select

Dentist list



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Dr. Emily Chen
Lead Hygienist & Treatment Coordinator

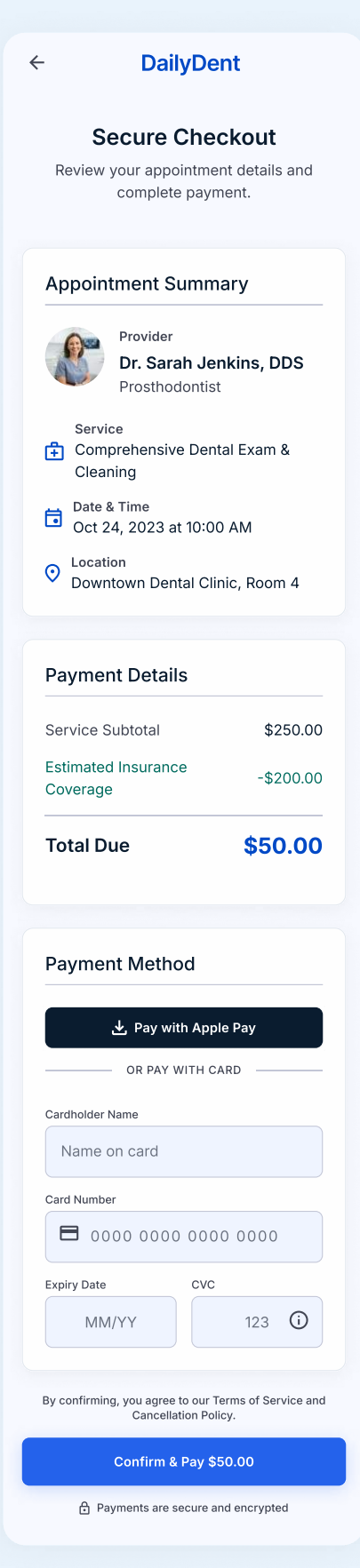
Welcome to your personalized care portal. Select a treatment below to review estimated timelines, costs, and clinical details before your upcoming consultation. Our goal is absolute transparency and your comfort.

Available Treatments

Preventative
Routine Checkup & Cleaning
A comprehensive examination of your oral health, including digital x-rays, professional plaque removal.
Duration: 45 mins
Est. Cost: $150 - $250
Select Treatment →

Cosmetic
Professional Whitening
Advanced, low-sensitivity clinical whitening treatment designed to lift deep stains and brighten your smile.
Duration: 60 mins
Est. Cost: $300 - $500
Select Treatment →

Dentist Info



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Secure Checkout
Review your appointment details and complete payment.

Appointment Summary
Provider: Dr. Sarah Jenkins, DDS, Prosthodontist
Service: Comprehensive Dental Exam & Cleaning
Date & Time: Oct 24, 2023 at 10:00 AM
Location: Downtown Dental Clinic, Room 4

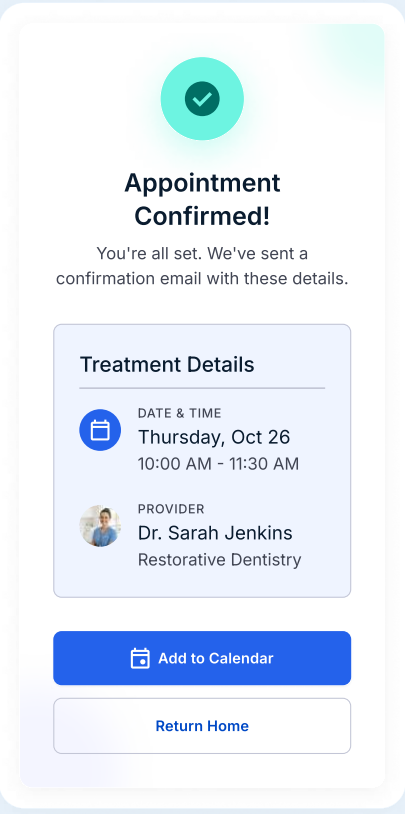
Payment Details

Service Subtotal	$250.00
Estimated Insurance Coverage	-$200.00
Total Due	$50.00

Payment Method
Pay with Apple Pay
OR PAY WITH CARD
Cardholder Name: Name on card
Card Number: 0000 0000 0000 0000
Expiry Date: MM/YY
CVC: 123

By confirming, you agree to our Terms of Service and Cancellation Policy.
Confirm & Pay $50.00
Payments are secure and encrypted

Check out

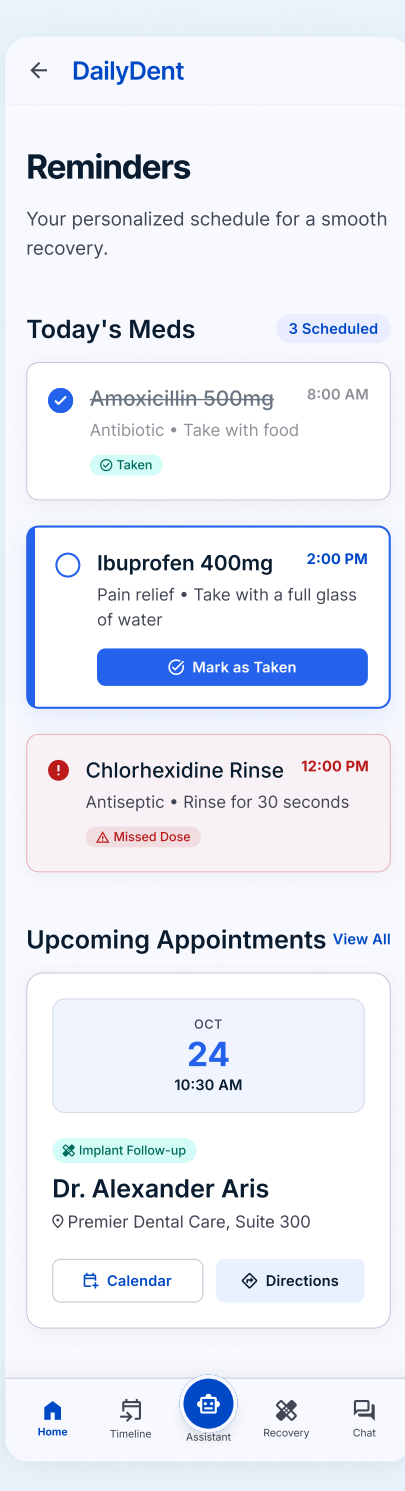


Appointment Confirmed!
You're all set. We've sent a confirmation email with these details.

Treatment Details
DATE & TIME: Thursday, Oct 26, 10:00 AM - 11:30 AM
PROVIDER: Dr. Sarah Jenkins, Restorative Dentistry

Add to Calendar
Return Home

Confirmation



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Reminders
Your personalized schedule for a smooth recovery.

Today's Meds — 3 Scheduled

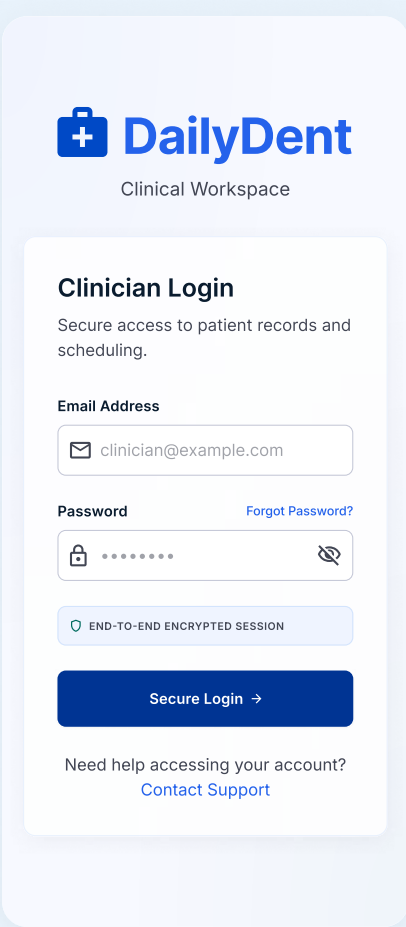
~~Amoxicillin 500mg~~ 8:00 AM
Antibiotic • Take with food
Taken

Ibuprofen 400mg 2:00 PM
Pain relief • Take with a full glass of water
Mark as Taken

Chlorhexidine Rinse 12:00 PM
Antiseptic • Rinse for 30 seconds
Missed Dose

Upcoming Appointments — View All
OCT 24 10:30 AM
Implant Follow-up
Dr. Alexander Aris
Premier Dental Care, Suite 300
Calendar
Directions

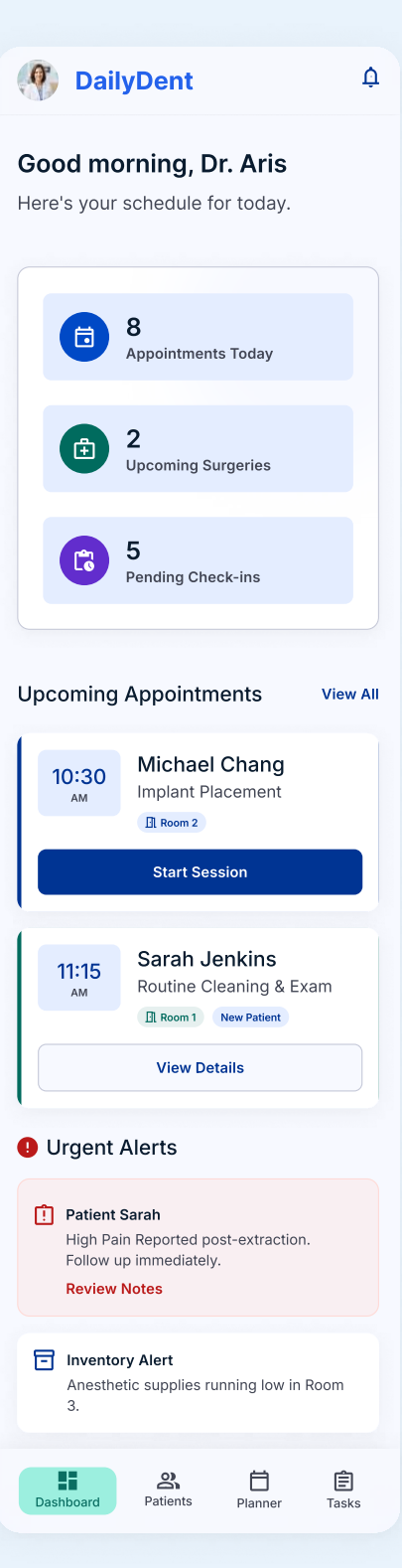
Confirmation



DailyDent
Clinical Workspace

Clinician Login
Secure access to patient records and scheduling.
Email Address: clinician@example.com
Password: ••••••••
Forgot Password?
END-TO-END ENCRYPTED SESSION
Secure Login →
Need help accessing your account? Contact Support

Dentist login



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Good morning, Dr. Aris
Here's your schedule for today.

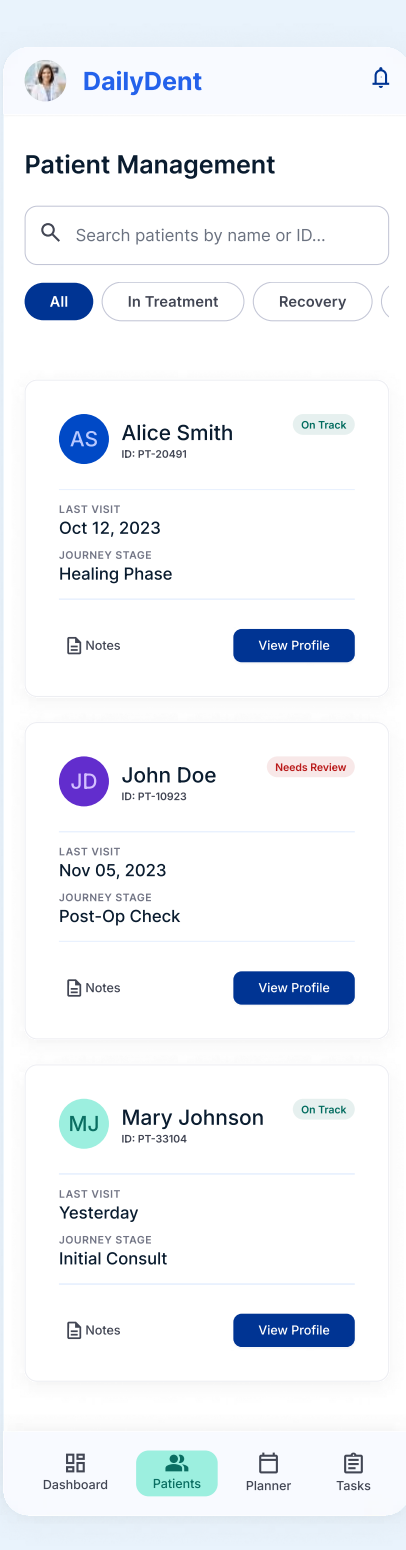
8 Appointments Today
2 Upcoming Surgeries
5 Pending Check-ins

Upcoming Appointments — View All
10:30 AM Michael Chang, Implant Placement, Room 2
Start Session
11:15 AM Sarah Jenkins, Routine Cleaning & Exam, Room 1, New Patient
View Details

Urgent Alerts
Patient Sarah
High Pain Reported post-extraction. Follow up immediately.
Review Notes

Inventory Alert
Anesthetic supplies running low in Room 3.

Dentist dashboard



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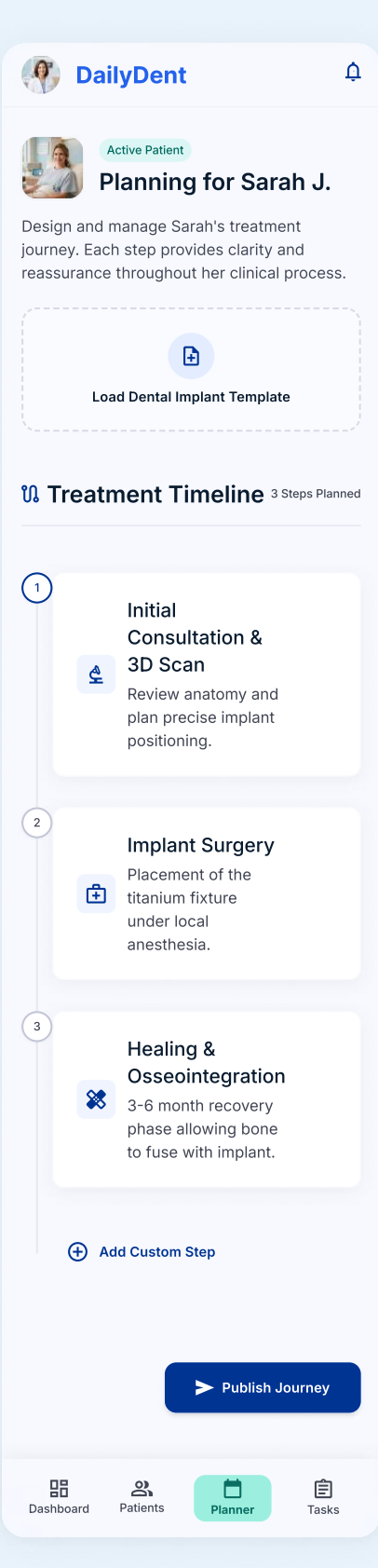
Patient Management
Search patients by name or ID...
All | In Treatment | Recovery

Patient	ID	Status	Last Visit	Journey Stage
Alice Smith	PT-20491	On Track	Oct 12, 2023	Healing Phase
John Doe	PT-10923	Needs Review	Nov 05, 2023	Post-Op Check
Mary Johnson	PT-33104	On Track	Yesterday	Initial Consult

Notes
View Profile

Patients info



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Active Patient
Planning for Sarah J.
Design and manage Sarah's treatment journey. Each step provides clarity and reassurance throughout her clinical process.

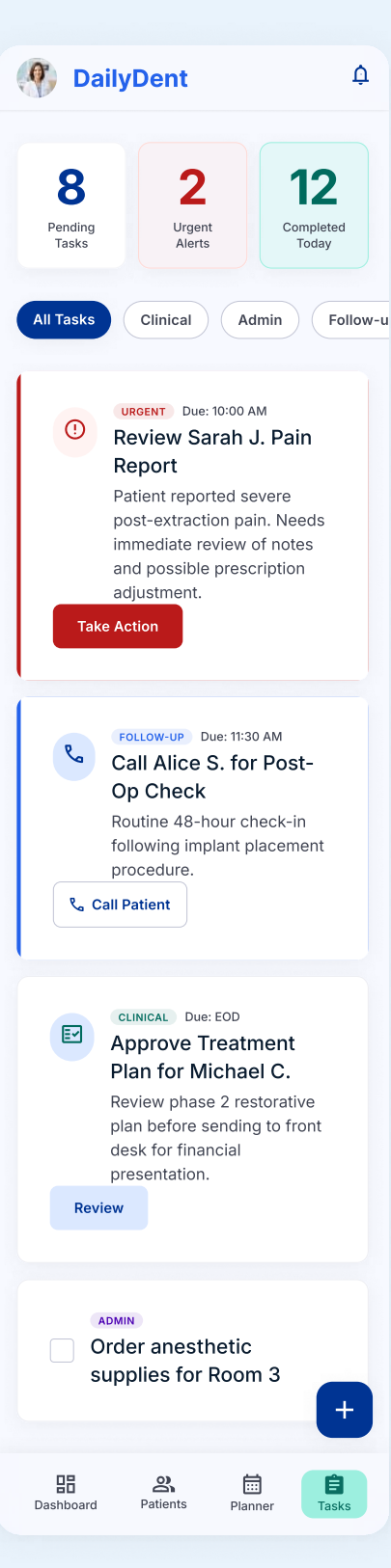
Load Dental Implant Template

Treatment Timeline — 3 Steps Planned

1. Initial Consultation & 3D Scan
Review anatomy and plan precise implant positioning.
2. Implant Surgery
Placement of the titanium fixture under local anesthesia.
3. Healing & Osseointegration
3-6 month recovery phase allowing bone to fuse with implant.

Add Custom Step
Publish Journey

Dentist planner



DailyDent

8 Pending Tasks
2 Urgent Alerts
12 Completed Today

All Tasks | Clinical | Admin | Follow-up

URGENT Due: 10:00 AM
Review Sarah J. Pain Report
Patient reported severe post-extraction pain. Needs immediate review of notes and possible prescription adjustment.
Take Action

FOLLOW-UP Due: 11:30 AM
Call Alice S. for Post-Op Check
Routine 48-hour check-in following implant placement procedure.
Call Patient

CLINICAL Due: EOD
Approve Treatment Plan for Michael C.
Review phase 2 restorative plan before sending to front desk for financial presentation.
Review

ADMIN
Order anesthetic supplies for Room 3

Dentist tasks

Thanks for Viewing